

BEJA, Portugal

WMO: 085620

Lat: 38.0259N

Lon: 7.8676W

Elev: 247

StdP: 98.40

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.3	3.6	-4.5	2.7	8.8	-2.2	3.2	9.4	10.4	12.4	9.3	12.3	2.7	290	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	37.5	20.1	35.7	19.7	33.9	19.1	21.7	32.6	21.0	31.5	20.3	30.3	3.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	22.8	18.3	13.6	22.1	17.7	13.0	21.6	64.2	32.7	61.5	31.4	59.3	30.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.7	6.8	0.1	40.9	1.4	1.9	-0.9	42.2	-1.7	43.3	-2.4	44.3	-3.4	45.7		
			-1.3	23.0	1.5	1.1	-2.3	23.8	-3.2	24.4	-4.0	25.0	-5.1	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	10.0	10.7	13.1	14.8	18.3	22.3	24.3	24.7	22.2	18.7	13.4	10.8
	DBStd	6.03	2.41	2.40	2.74	3.13	3.73	3.57	2.97	2.91	2.80	3.05	2.84	2.47
	HDD10.0	78	29	19	5	1	0	0	0	0	0	0	4	19
	HDD18.3	1216	257	213	163	114	48	6	0	0	2	31	148	233
	CDD10.0	2631	30	39	101	145	256	371	442	457	367	271	107	45
	CDD18.3	727	0	0	1	8	46	126	184	198	119	44	1	0
	CDH23.3	8319	0	0	16	102	583	1571	2288	2310	1136	311	2	0
	CDH26.7	4131	0	0	1	16	226	797	1254	1254	493	91	0	0
Wind	WSAvg	3.3	3.2	3.4	3.4	3.5	3.4	3.4	3.7	3.5	3.1	3.1	3.1	3.1
Precipitation	PrecAvg	583	80	69	51	58	43	21	2	3	25	66	76	88
	PrecMax	1042	275	164	143	152	149	84	19	38	118	214	283	279
	PrecMin	355	0	3	0	4	0	0	0	0	1	0	2	4
	PrecStd	156	64	52	38	38	37	22	4	8	27	54	59	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.9	25.2	28.6	34.4	38.7	39.8	39.5	36.4	32.2	23.2	18.9
		MCWB	12.9	13.8	15.1	16.7	18.5	19.8	20.2	21.0	19.5	17.3	16.1	13.9
	2%	DB	16.2	18.0	23.1	26.4	31.7	36.1	37.5	37.2	33.7	29.2	21.1	17.0
		MCWB	12.6	12.9	14.4	16.6	17.8	19.6	20.0	20.2	18.6	17.1	15.6	14.3
	5%	DB	15.0	16.6	21.0	24.0	29.2	33.9	35.4	35.4	31.7	26.7	19.5	15.9
		MCWB	12.5	12.3	14.0	15.4	17.2	19.2	19.5	19.8	18.3	17.0	15.0	13.7
	10%	DB	14.0	15.2	18.9	21.6	26.7	31.4	33.3	33.2	29.6	24.5	18.0	14.9
		MCWB	12.0	11.9	13.3	14.6	16.6	18.5	19.2	19.2	18.2	16.7	14.7	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.6	17.3	19.0	20.9	22.0	22.4	23.0	21.6	20.4	18.5	16.1
		MCDB	16.2	17.5	21.9	25.8	30.3	33.9	34.0	34.5	30.7	24.7	20.6	16.9
	2%	WB	14.1	14.5	15.8	17.4	19.4	20.9	21.5	21.7	20.6	19.4	17.3	15.2
		MCDB	15.0	16.3	20.2	23.9	27.9	32.4	32.9	32.7	28.7	23.7	19.1	16.1
	5%	WB	13.3	13.5	14.8	16.2	18.3	20.0	20.7	21.0	19.9	18.6	16.3	14.2
		MCDB	14.4	15.4	18.9	22.2	25.9	30.8	32.0	31.6	27.1	22.7	18.1	15.4
	10%	WB	12.4	12.6	13.9	15.1	17.2	19.2	20.0	20.2	19.2	17.9	15.4	13.3
		MCDB	13.6	14.5	17.7	20.4	24.7	29.4	31.2	30.3	25.9	21.8	17.2	14.5
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.8	10.1	11.1	13.0	15.4	16.8	16.1	13.2	10.1	8.1	7.5
		MCDBR	8.2	10.4	13.3	14.5	17.0	18.8	19.4	18.3	16.4	13.5	9.4	7.9
		MCWBR	5.5	6.3	6.6	6.5	6.3	6.0	5.9	5.7	5.4	5.1	4.9	5.1
	5% WB	MCDBR	6.4	8.5	11.0	12.8	14.8	17.0	17.6	16.4	13.0	9.7	7.1	6.5
		MCWBR	5.1	5.8	6.4	6.6	6.5	6.3	6.1	5.9	5.2	4.8	4.6	4.9
Clear-Sky Solar Irradiance	taub	0.313	0.329	0.361	0.367	0.374	0.378	0.357	0.375	0.372	0.351	0.326	0.315	
	taud	2.500	2.453	2.370	2.366	2.370	2.374	2.406	2.368	2.400	2.484	2.512	2.507	
	Ebn at Noon	872	900	899	912	906	899	915	892	876	864	848	842	
	Edh at Noon	79	94	113	120	122	121	116	118	108	90	78	74	
All-Sky Solar Radiation	RadAvg	2.41	3.35	4.48	5.62	6.76	7.63	7.90	7.00	5.40	3.76	2.66	2.15	
	RadStd	0.30	0.40	0.47	0.48	0.56	0.40	0.21	0.22	0.34	0.31	0.30	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page